

Somaiya Vidyavihar University
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Batch: A-4 **Roll No.:** 16010122151

Experiment / Assignment / Tutorial No: 5

Title: M/M/1 and M/G/1 Queuing Model Simulation

Objective: The objective of this lab experiment is to perform an analysis of the M/M/1 and M/G/1 queue model by considering different varying parameters and their impact on key performance metrics. The experiment includes the calculation of theoretical values, simulation of the queue, and statistical analysis of the results.

Expected Outcome of Experiment:

CO2: Analyse and apply general principles of event scheduling algorithm & various statistical methods on different applications.

Books/ Journals/ Websites referred:

1. “Discrete-Event System Simulation” by Jerry Banks, John S. Carson II, Barry L. Nelson, David M. Nicol.
 2. SimPy Documentation: <https://simpy.readthedocs.io/en/latest/>
 3. SciPy Documentation: <https://docs.scipy.org/doc/scipy/>
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Background:

(Explain in brief M/M/1 and M/G/1 with key performance metrics)

Queueing theory provides mathematical models for analyzing waiting lines, which are ubiquitous in computer networks, telecommunications, and service systems. This lab focuses on simulating and analyzing two fundamental single-server queueing models: M/M/1 and M/G/1.

1. M/M/1 Queue Model

The M/M/1 queue is the simplest and most well-known queueing model. Its notation is defined as follows:

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- The first M stands for "Memoryless" and denotes that the arrival process is a Poisson process. This means customers arrive randomly and independently at a constant average rate (λ).
- The second M also stands for "Memoryless" and indicates that the service times follow an exponential distribution with a constant average service rate (μ). The "memoryless" property implies that the probability of a service being completed is independent of how long it has already been in progress.
- The 1 indicates a single server.

The key performance metric for this system is the Traffic Intensity or Utilization (ρ), given by $\rho = \lambda / \mu$. For the queue to be stable (not infinitely long), ρ must be less than 1. Other key metrics derived from theory include:

- L: The average number of customers in the system (both waiting and in service).
- W_q : The average time a customer spends waiting in the queue before being served.
- The distributions of queue length and waiting time, which are geometric and exponential, respectively.

2. M/G/1 Queue Model

The M/G/1 model is a generalization that provides more flexibility for modeling real-world systems. Its notation is:

- M: The arrival process remains Markovian (Poisson) with rate λ .
- G: The service time distribution is General (i.e., it can be any distribution with a known mean and variance). This is crucial for scenarios where service times are not memoryless, such as fixed-duration tasks or normally distributed processing times.
- 1: A single server.

The key difference from the M/M/1 model is that the performance metrics now depend on the variance of the service time distribution, not just its mean. The stability condition remains $\rho = \lambda * E[S] < 1$, where $E[S]$ is the mean service time. The average waiting time, for instance, is given by the Pollaczek-Khinchin (P-K) formula, which explicitly includes the service time variance. This model is better suited for systems like a toll booth where service times may be relatively constant or follow a different pattern altogether.

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Problem Statement 1:

Perform analysis of the M/M/1 queue model by considering different levels of traffic intensity and their impact on key performance metrics.

Consider the following:

Traffic Intensity (ρ) Levels:

Low Traffic Intensity: $\rho=0.5$

Moderate Traffic Intensity: $\rho=0.75$

High Traffic Intensity: $\rho=0.95$

Parameters:

Arrival rate (λ): Adjusted based on the chosen ρ

Service rate (μ): Fixed at 1 customer per minute

Simulation time: 100,000 minutes

Key Performance Metrics

Average Waiting Time (W_q)

Average Number of Customers in the System (L)

Server Utilization (ρ)

Queue Length Distribution

Waiting Time Distribution

Problem Statement 2:

- Simulate an M/G/1 queue to model peak-hour traffic at a toll booth, where service times vary throughout the day, and analyze the toll booth's performance metrics.
- Vehicles arrive following a Poisson process with an average arrival rate.
- Service times vary according to a normal distribution with different means during peak and off-peak hours.
- Analyze the average waiting time, queue length, and server utilization during peak and off-peak hours.

Key Performance Metrics:

Average Waiting Time (W_q)

Average Number of Customers in the System (L)

Server Utilization (ρ)

Consider the following:

1. **Arrival Rate (λ):** Average 5 vehicles per minute.
2. **Service Time Distribution:**
 - Peak Hours: Mean = 0.5 minutes, Standard Deviation = 0.1 minutes.
 - Off-Peak Hours: Mean = 1.0 minutes, Standard Deviation = 0.2 minutes.
3. **Peak Hours Duration:** 7:00 AM - 9:00 AM.
4. **Off-Peak Hours Duration:** 9:00 AM - 5:00 PM.

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5. **Simulation Time:** 10 hours (7:00 AM - 5:00 PM).

Simulation:

Implement the simulation using Python and the SimPy library to model the queue behavior. The key steps include defining the customer arrival process, handling the customer service process, and collecting statistics on waiting times and queue lengths.

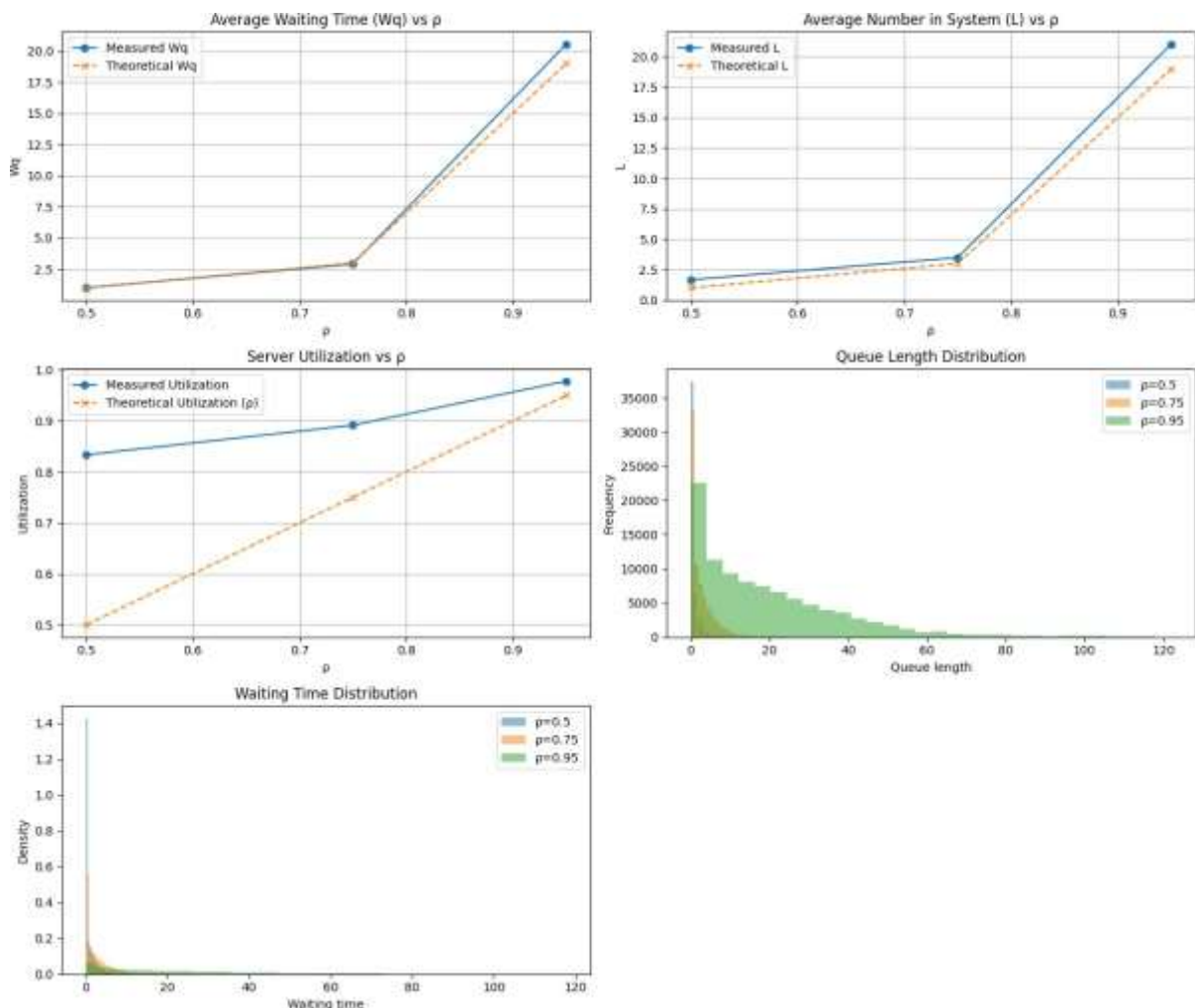
Implementation Steps with Screen shots:

Problem Statement -1 : code: attached - [expt5-1.py](#)

Problem Statement -2 : code: attached - [expt5-2.py](#)

Output(s) :

Problem Statement-1:



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Problem Statement-2 :

Simulating M/G/1 Toll Booth Queue with varying service times...

--- Simulation Results for 10 Hours ---

Arrival Rate (λ): 5 vehicles/minute

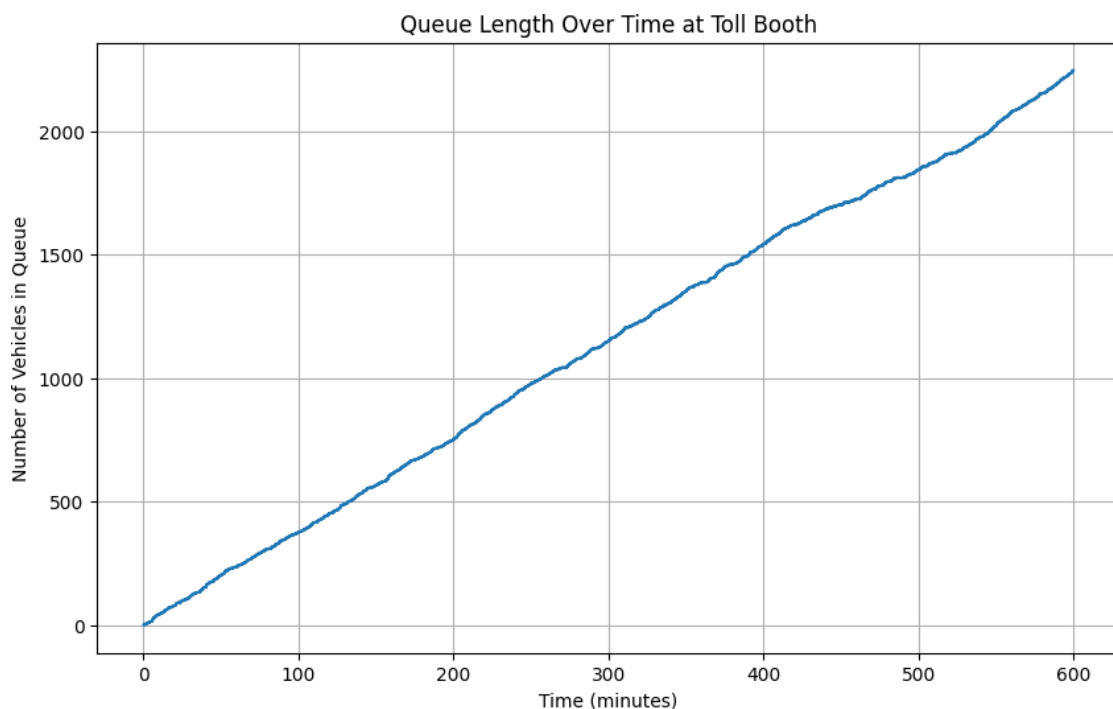
Peak Hours Service Mean: 0.5 min, Std: 0.1 min

Off-Peak Hours Service Mean: 1.0 min, Std: 0.2 min

Average Waiting Time (W_q): 255.6867 minutes

Average Number of Customers in System (L): 256.2552

Estimated Server Utilization (ρ): 0.9984



Conclusion:

This analysis demonstrates that key performance metrics in queueing systems, such as waiting time and queue length, are highly sensitive to traffic intensity. Furthermore, the M/G/1 model confirms that increased variance in service times significantly degrades system performance compared to the memoryless service assumption of the M/M/1 model.

Post lab Questions:

Network of Queues (M/M/1 and M/M/3)

Simulate a network of interconnected queues (M/M/1 and M/M/3) where customers pass through multiple service stations with different service rates.

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Consider the following Scenario:

- Customers first arrive at a check-in counter (M/M/1).
- After check-in, customers move to a service desk (M/M/1).
- After the service desk, customers may visit one of several specialized service counters (M/M/3).
- Each queue has different arrival and service rates.
- Analyze the overall system performance, including average waiting time, queue length, and server utilization at each station.

CODE: Attached with submission

OUTPUT:

```
C:\Users\Soumil\Downloads>python exp5(pl).py
Entry Gate Statistics:
Mean Delay: 0.33 units
Mean Line Length: 0.13 items
Usage Factor: 0.27

Inspection Gate Statistics:
Mean Delay: 0.00 units
Mean Line Length: 0.00 items
Usage Factor: 0.35

Final Gate Statistics:
Mean Delay: 0.00 units
Mean Line Length: 0.00 items
Usage Factor: 0.15
```